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Evaluating New LinkedIn Labor Market Tightness Indicators

Elizabeth B. Li, Giorgio Topa, and Jacob P. Weber



Measures of labor market tightness (LMT) describe how difficult it is for firms to find workers. The labor market is considered tight if there are few job seekers available relative to job openings, implying stronger wage pressures as employers compete for scarce job seekers. Thus, good measures of LMT are important for predicting wage inflation and calibrating monetary policy. Recently, the hiring platform LinkedIn began producing three innovative LMT measures based on its user data. In this post, we use public data from the Job Openings and Labor Turnover Survey (JOLTS) and the Bureau of Labor Statistics (BLS) to evaluate the ability of these novel measures to predict wage and price inflation relative to existing LMT measures, finding that one stands out as particularly useful and timely.

How Are the New LinkedIn Measures Constructed?

We study three LinkedIn measures of LMT: the Jobs per Application Index (JPAI), the Jobs per Jobseeker Index (JPJI), as defined by McCrory and Huang (2025), and the LinkedIn Hiring Rate (LHR), as defined by Chua and Kantenga (2024) and first released in 2017.

The JPAI is defined as the number of attractive job openings in a given month divided by the number of job applications submitted on LinkedIn. An attractive job opening on LinkedIn is defined as satisfying two conditions: the job poster either paid to promote the role or manually listed the position on LinkedIn, and thus paid the opportunity cost of their time, and, within a period of time t, the job listing received an application from a LinkedIn member in the labor market where the job is located. Here, an application is either submitted directly via LinkedIn's Easy Apply feature or inferred because the member clicked on a link to apply to the role outside of LinkedIn's platform.

The JPJI is defined as the number of attractive job openings that month divided by the number of jobseekers that submitted at least one job application on LinkedIn that month. An attractive job and a job application are both defined in the same way as for the JPAI.

Finally, the LHR is defined as the number of LinkedIn members who added a new employer to their profile divided by the relevant LinkedIn member base (for example, members located in the U.S.). This number is indexed to the average month in 2016; for example, an index of 1.05 indicates a hiring rate that is 5 percent higher than the average month in 2016. In our analysis we use the seasonally adjusted data for the LHR, as produced by LinkedIn.

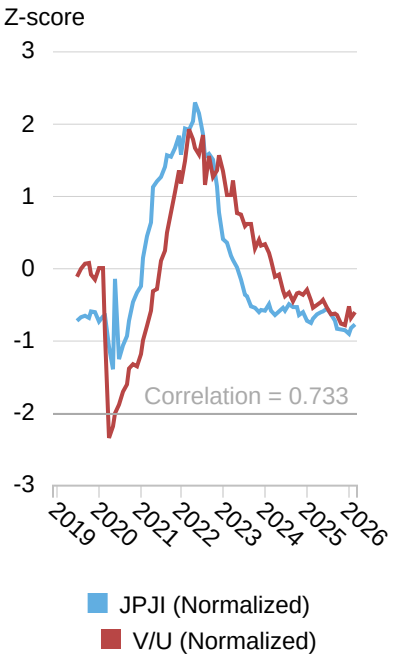
All three monthly measures have the advantage of being timelier than standard measures. For instance, data for August 2026 could be made available in the first week of September, much sooner than JOLTS data which are available with about a five-week lag. However, these measures are not necessarily representative of the U.S. economy as a whole, since they capture only jobs posted and applied to on LinkedIn, which raises the following question: how correlated are they with existing measures of LMT, and are they useful for predicting wage and price inflation?

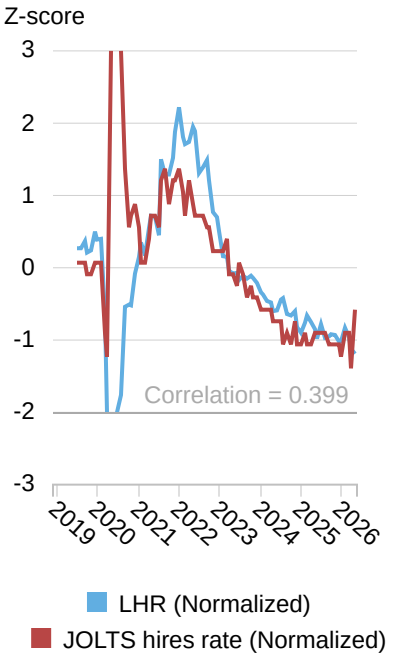
Comparing New and Existing Measures

How do the LinkedIn LMT measures compare to their more widely known counterparts? As a benchmark, we use the vacancies-to-unemployment (V/U) ratio, a widely used measure of labor market tightness defined as the number of job vacancies (the number of job openings posted by employers each month as measured by JOLTS) divided by the number of people who are unemployed. We also benchmark the LHR with the hires rate from JOLTS. In all analyses, we restrict attention to the period starting in July 2019 (given limited historical data availability for the JPJI and the JPAI) and ending in April 2026.

The left panel of the chart below shows that the JPJI tracks V/U well. The correlation between the two measures is 0.733. The JPJI is slightly higher than V/U in the early aftermath of the COVID-19 pandemic, and slightly lower from late 2022 to mid-2024. The right panel of the chart shows that the LHR and the JOLTS hires rate are also very closely linked; when starting the data series from the beginning of 2021 (excluding the large movements that occurred at the onset of COVID-19), the correlation between the two (normalized) variables is 0.95.

The JPJI and V/U Track Closely over Time, As Do the LHR and the JOLTS Hires Rate





Sources: LinkedIn; Job Openings and Labor Turnover Survey (JOLTS); Bureau of Labor Statistics.
Notes: All variables are normalized as Z-scores over the plotted time period (centered around their mean and divided by their standard deviation). The JPJI is the Jobs per Jobseeker Index, V/U is the vacancies-to-unemployment ratio, and LHR is the LinkedIn Hiring Rate.

How Good Are the New Measures for Nowcasting and Forecasting?

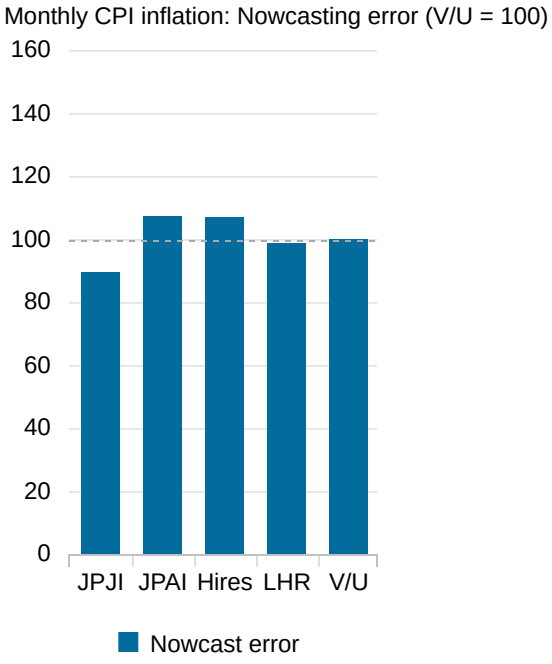
We now examine the LinkedIn measures' nowcasting and forecasting properties in predicting price and wage inflation in real time. For these exercises, we use monthly Consumer Price Index (CPI) data and quarterly Employment Cost Index (ECI) data, respectively. In the nowcasting exercise, we assess the ability of LMT measures to predict the current values of the CPI and the ECI. Nowcasting is useful for

helping policymakers to predict inflation before the official data are released. In practice, we run rolling regressions of inflation up to period *t-1* (the last period official data are available) on LMT measures. We then use the predicted inflation in period *t* (using LMT measures through *t*) as our nowcast.

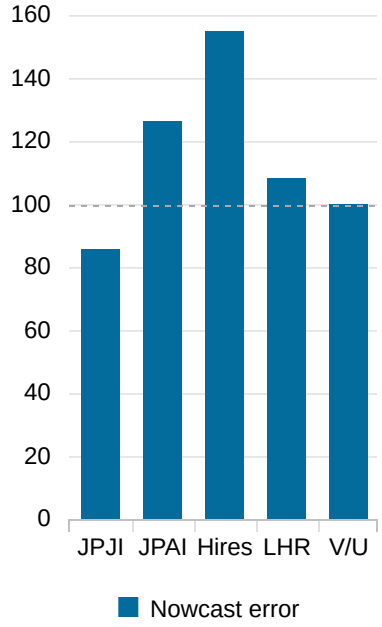
How do we evaluate the "goodness of fit" of our nowcast measures? We use the root mean squared error (RMSE) as our metric for how well the nowcast matches actual inflation. After nowcasting values for CPI and ECI inflation, we compute the RMSE for how accurately each LMT-predicted inflation measure tracks actual inflation. We then take the average RMSE of each measure and compare how they fare against the average RMSE of V/U as a benchmark.

The chart below shows that the JPJI and the LHR are consistently the best performers. The JPJI has the lowest nowcasting error, performing better than V/U in predicting both price and wage inflation. The LHR performs better than V/U for CPI but worse for ECI. The JPAI and the JOLTS hires rate are consistently worse than V/U.

The JPJI Has the Best Nowcasting Performance in Predicting CPI and ECI Inflation



Quarterly ECI inflation: Nowcasting error (V/U = 100)



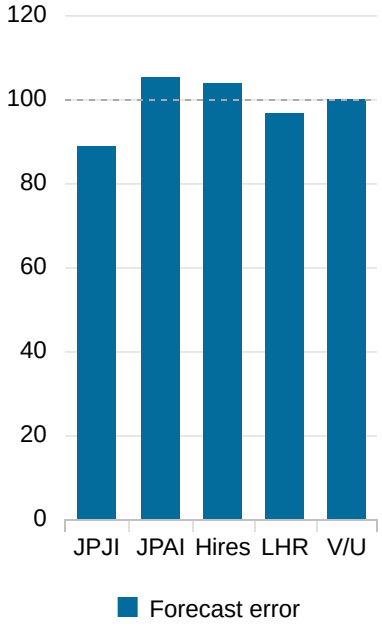
Sources: LinkedIn; Job Openings and Labor Turnover Survey (JOLTS); Bureau of Labor Statistics.
Notes: Nowcast errors involve comparing each measure's mean root mean squared error (RMSE) relative to the vacancies-to-unemployment (V/U) ratio's mean RMSE. Consumer Price Index (CPI) inflation is a monthly measure, while Employment Cost Index (ECI) inflation is quarterly. Hires refers to the JOLTS hires rate while the Jobs per Jobseeker Index (JPJI), the Jobs per Application Index (JPAI), and the LinkedIn Hiring Rate (LHR) are from LinkedIn.

We conduct the same exercise for forecasting. Here we regress each inflation measure on one-period lagged values of our LMT measures through period *t*, to construct a forecast for *t+1* inflation. Similarly to what we did with our nowcasting exercise, we evaluate the effectiveness of the new LinkedIn measures by computing the RMSE for how accurately each LMT-predicted inflation measure tracks actual inflation. We then take the average RMSE of each measure and compare how they fare against the average RMSE of V/U.

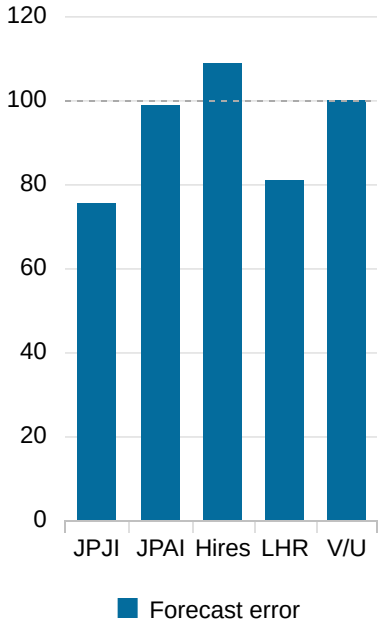
The chart below shows that, as with nowcasting, the JPJI is the best performing measure, with a lower forecasting error than V/U in predicting both price and wage inflation. Notably, the LHR also performs better than V/U in predicting both inflation measures, though still worse than the JPJI. And again, the JPAI and the JOLTS hires rate are the worst-performing measures, though the JPAI performs better than V/U in forecasting ECI inflation.

The JPJI Also Has the Best Forecasting Performance for CPI and ECI Inflation

Monthly CPI inflation: Forecasting error (V/U = 100)



Quarterly ECI inflation: Forecasting error (V/U) = 100



Sources: LinkedIn; Job Openings and Labor Turnover Survey (JOLTS); Bureau of Labor Statistics.
Notes: Forecast errors involve comparing each measure's mean root mean squared error (RMSE) relative to the vacancies-to-unemployment (V/U) ratio's mean RMSE. Consumer Price Index (CPI) inflation is a monthly measure, while Employment Cost Index (ECI) inflation is quarterly. Hires refers to the JOLTS hires rate while the Jobs per Jobseeker Index (JPJI), the Jobs per Application Index (JPAI), and the LinkedIn Hiring Rate (LHR) are from LinkedIn.

Conclusion

In this post, we have investigated the performance of three novel measures of labor market tightness produced by LinkedIn in predicting wage and price inflation. We find that the JPJI and the LHR not

only track existing measures of LMT well, but also have a RMSE that is comparable to or better than existing, trusted LMT measures like V/U in nowcasting and forecasting price and wage inflation. Crucially, these measures are also timelier than the JOLTS hires rate and traditional measures of tightness, such as V/U.

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